

TECHNICAL INVESTIGATION REPORT

Investigating Over-Quantization Effects

Analyzing Dynamic Range Collapse, Model Degradation, and Mitigation Strategies in
Quantized Neural Networks & Signals

PREPARED FOR
AI Research & Systems Architecture

DATE
July 2026

FORMAT
Executive Presentation Deck

What is Over-Quantization?

Definition & Thresholds

Quantization maps high-precision continuous values (e.g., FP32/FP16) to discrete lower-precision representations (e.g., INT8, INT4, INT2). **Over-quantization** occurs when bit-width reduction crosses a critical entropy threshold, causing unrecoverable loss of variance, resolution, and information capacity.

Primary Mathematical Drivers

- **Clipping Error:** Truncating tail values outside dynamic range bound $[-\alpha, \alpha]$.
- **Rounding Error:** Step size resolution grid becomes too coarse to capture subtle gradients.
- **Variance Collapse:** Dense feature representations collapse into zero-gradient step functions.

The Precision Continuum

Precision	Dynamic Range	Status / Risk
FP32 / FP16	Continuous ($\sim 10^{38}$)	Ideal Quality
INT8	256 Discrete Levels	Optimal Standard
INT4 (AWQ/GPTQ)	16 Discrete Levels	Managed Risk
INT2 / INT1	2–4 Discrete Levels	Over-Quantized

Critical Insight

Unlike linear degradation, over-quantization exhibits a sharp non-linear performance cliff. Model performance remains robust down to INT4, but rapidly collapses near 2-bit quantization.

Visualizing Information & Weight Collapse

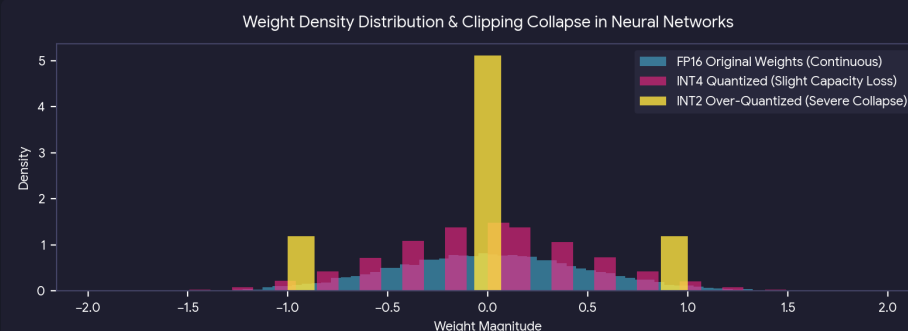
1. Signal & Image Banding Artifacts

Visualizing smooth continuous color space reduction into quantized steps. Extreme quantization (1-bit) completely eliminates subtle spatial transitions, creating false edges and posterization.



2. Neural Network Weight Collapse

In LLM compression, pushing below INT4 causes weight distributions to collapse into discrete spikes, completely destroying subtle activation dynamics and cross-layer attention.



Key Limitations & Failure Modes

1. LLMs & GenAI

- **Perplexity Explosion:** Exponential jump in token generation error rates below 3-bit.
- **Outlier Sensitivity:** Key feature activations (salient weights) clipped, destroying reasoning capabilities.
- **Repetitive Loops:** Degenerate sampling dynamics and endless repeating phrases.

2. Vision & Perception

- **Posterization & Banding:** False contouring in continuous gradient backgrounds.
- **False Boundary Detection:** Edge detection algorithms misinterpret quantization boundaries as real object edges.
- **Small Object Disappearance:** Low-amplitude high-frequency details zeroed out completely.

3. Signal & Audio

- **Harmonic Distortion:** Quantization noise becomes correlated with input signal.
- **High-Frequency Loss:** Fine transient details and room acoustics flattened.
- **SNR Degradation:** ~6 dB loss in Signal-to-Noise Ratio for every bit dropped (SQNR rule).

Mitigating Over-Quantization Artifacts

1. Outlier-Aware Quantization (AWQ & SmoothQuant)

Protects the ~1% critical activation channels by keeping them in higher precision (FP16/INT8) while quantizing 99% of remaining weights to low-bit targets, avoiding global accuracy drop.

2. Mixed-Precision Architectures

Assigning variable bit-widths per layer based on sensitivity analysis (Hessian spectrum). Sensitive layers (first/last layers, cross-attention) stay at INT8, less sensitive layers drop to INT4.

3. Quantization-Aware Training (QAT)

Simulates quantization rounding/clipping during forward-pass fine-tuning via straight-through estimators (STE), allowing weights to adapt to discrete boundaries prior to deployment.

4. Stochastic Dithering (Signals/Vision)

Injecting low-level pseudorandom noise prior to quantization to break deterministic step banding, converting harsh visual/audio harmonic distortion into neutral wideband noise.

Key Takeaways & Deployment Rules

Target Bit-Width	Recommended Use Case	Risk Level
8-Bit (INT8)	Production Default, Zero-loss inference	Negligible Risk
4-Bit (AWQ/ GPTQ)	Edge LLMs, High-throughput serving	Low Risk (with Outlier protection)
3-Bit (Grouped)	Extreme memory-constrained environments	Moderate Risk
2-Bit / Binarized	Research / Specialized binary networks	High Over-Quantization Risk

Architectural Golden Rule

Never apply uniform low-bit (< 4-bit) quantization across all layers without outlier preservation mechanisms or fine-grained per-channel grouping.

Summary Recommendations

- Establish Entropy Baseline:** Profile target model layers using Hessian-based sensitivity checks before selecting bit-widths.
- Adopt AWQ / SmoothQuant:** Ensure dynamic feature outliers are protected during quantization.
- Monitor Perplexity Cliffs:** Conduct rigorous evaluation on long-context and edge-case evaluation datasets.